nScout

Network Traffic Analysis & Security Monitoring

Document purpose: public-facing technical overview and practical orientation; not a security certification or independent audit.

Project status: Active development; v0.2.0 stable and v0.3 milestone on main

Technology / architecture: Python, FastAPI, Scapy, React, WebSockets, MongoDB, PyInstaller, Inno Setup

Architecture & capabilities

• Live packet capture, imported PCAP analysis and simulated traffic

• Packet inspection: Ethernet, ARP, IPv4/IPv6, TCP, UDP, ICMP, DNS, visible HTTP and observable TLS metadata

• Connection stories, TCP health, host inventory, network map, timeline and Boolean search

• Heuristic findings with severity, confidence and supporting evidence

• Investigation exports in JSON, standalone HTML and PDF

Evaluation / getting started

• Windows: obtain current stable installer from Releases, install and launch

• Use PCAP mode for offline evaluation; Npcap and suitable permissions required for live Windows capture

• Use only authorized captures; preserve the original PCAP and record version, filters and findings

Limitations and security considerations

• Heuristic detections are investigation leads, not proof of compromise

• HTTPS application payloads are not decrypted

• Windows-native capture without Npcap is a research goal, not a completed replacement

Engineering roadmap

• Expand regression PCAPs and performance profiling

• Validate desktop lifecycle and explore WFP capture parity

Source and verification

Repository: https://github.com/ibrahim1101/nScout

Information reflects the accessible project README and the supplied engineering resume. Features may change; verify current repository branches, releases and documentation before deployment. Do not upload private keys, user data, internal logs or unredacted evidence.